

Event Time Dilation and the Variance Clock

Two clocks, day-weighted events, and a price-preserving vol drop

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Abstract

Variance does not accrue uniformly in calendar time: a scheduled event — earnings, an FOMC print, a court ruling — packs the variance of several normal days onto one. This note documents the Vol-Fitter’s *variance clock*, which makes that explicit by carrying two times: calendar time t , and a dilated variance time τ in which an event day counts as several. Each event adds a number of *extra equivalent days*; the surface is fitted and quoted in weighted years $\tau = \tau_{\text{days}}/365$. The key invariant is price-preservation: total variance $w(k)$ is fixed by the observed option price and is clock-independent, so the working implied volatility $\sqrt{w/\tau}$ *drops* when an event is inserted before the expiry — exactly the overnight-vol-crush traders expect. An optional normalization rescales all days so the one-year variance budget is unchanged, making events *redistribute* variance within the year rather than add it. The calendar can be typed in or *auto-calibrated*: a solver places events so the event-time forward variance is as flat as possible, inferring the calendar from the term structure’s kinks, after which it is editable. With no events the variance clock collapses to calendar time and every downstream fit is byte-identical to the plain pipeline. A worked example shows a 4-extra-day earnings event lifting short-dated implied vol and decaying away, with a working-vol drop of 36 vol bps at a fixed expiry just past the event.

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1 Why two clocks

A diffusion model assumes variance accrues at a steady rate: total variance $w = \sigma^2 t$ is linear in calendar time. Real markets violate this around scheduled events, where a single day carries the variance of many. Forcing such a surface onto a single calendar clock produces the familiar pathology — a short-dated implied vol that looks “too high” and a term structure with a kink that a smooth model cannot represent without distorting the smile.

Heuristic

Separate the *calendar* (how many days until expiry) from the *variance budget* (how much variance those days carry). An earnings day is one calendar day but, say, five days’ worth of variance. If we measure time in *variance days*, the diffusion looks steady again — and the “too high” short-dated implied vol is just the same total variance divided by a smaller calendar time. The event clock is the change of variable that restores $w = \sigma^2 \tau$ with a constant σ .

Invariant

1. **Price preservation.** The clock never touches $w(k)$; it changes only the volatility read-off $\sqrt{w/\tau}$. Adding an event cannot change an option price, hence cannot introduce arbitrage.
2. With an empty calendar, $\tau = t$ exactly and the entire downstream pipeline is byte-identical.
3. Every consumer — every fit, the LV surface, the term structure, the table — reads the *same* τ ; there is one variance clock per node, not one per view.
4. Editing the calendar refits only the affected ticker (per-ticker events version).

2 The variance clock

2.1 Day-weighted variance time

Each event carries a number N_e of *extra* equivalent days (an earnings day with $N_e = 4$ counts as 5 normal days of variance). The variance time to a calendar maturity t is

$$\tau_{\text{days}}(t) = t \cdot 365 + \sum_{t_e \leq t} N_e, \quad \tau(t) = \frac{\tau_{\text{days}}(t)}{365}, \quad (1)$$

counting only events at or before t . Figure 1 plots $\tau(t)$: it tracks the calendar line until the event, jumps by $N_e/365$ there, and runs parallel afterward. The fits and quotes live on τ ; with no events, $\tau(t) = t$ exactly. The clock (1), with the optional 1Y-budget normalization, is direct (matches production to 10^{-12}):

Listing 1: The event variance clock $\tau(t)$ (1) (distilled from `calib/weighted_time.py::weighted_variance_years`).

```
1 def weighted_variance_years(t_cal, events, normalize=False, dpy=365.0)
  :
2     if t_cal <= 0: return t_cal          # events: (t_e, N_e) pairs,
      N_e = extra days
```

```

3     tau_days = t_cal*dpy + sum(N for te, N in events if te <= t_cal
4         and N > 0)
5     if normalize:                                # rescale so a 1Y budget
6         stays dpy days
7         e1 = sum(N for te, N in events if te <= 1.0 and N > 0)
8         tau_days *= dpy/(dpy + e1)
9     return tau_days/dpy                          # tau(t); equals t_cal with
10         no events

```

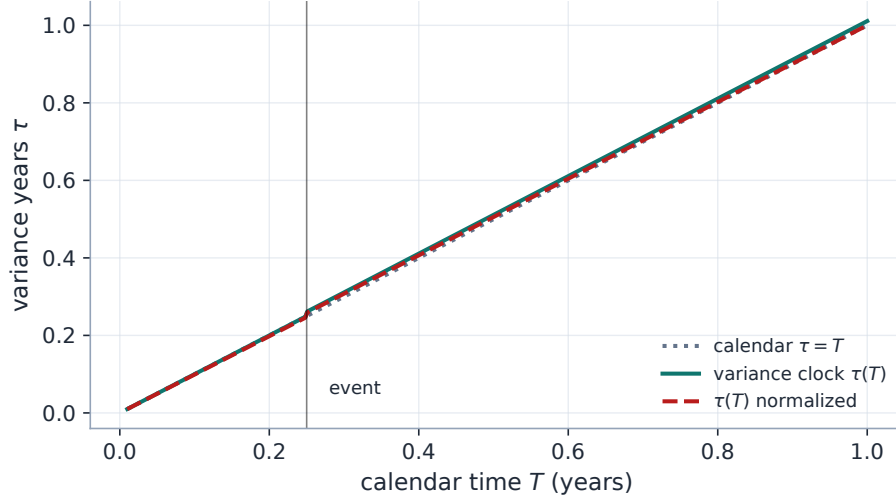


Figure 1: Variance time $\tau(t)$ for an event at $t_e = 0.25$ worth 4 extra days: it jumps at the event and runs parallel to calendar time thereafter. The normalized clock (dashed) rescales so the one-year budget is unchanged.

2.2 Price preservation

The crucial invariant is that the *total variance* $w(k)$ is fixed by the observed option price — it is the Black total variance that reprices the quote, and that is clock-independent. The variance clock only changes how we *report* it as a volatility:

$$\sigma_{\text{work}}(k) = \sqrt{\frac{w(k)}{\tau}}. \quad (2)$$

Inserting an event before the expiry raises τ at fixed w , so the working implied vol *drops*. This is the vol crush: the price is unchanged, but the *diffusive* volatility implied by it is lower once the event's variance is accounted for separately. Figure 2 shows the term-structure consequence: with the event, short-dated calendar implied vol is lifted (the event's variance compressed into few calendar days) and decays as the event becomes a smaller fraction of the maturity; without it, the term structure is flat.

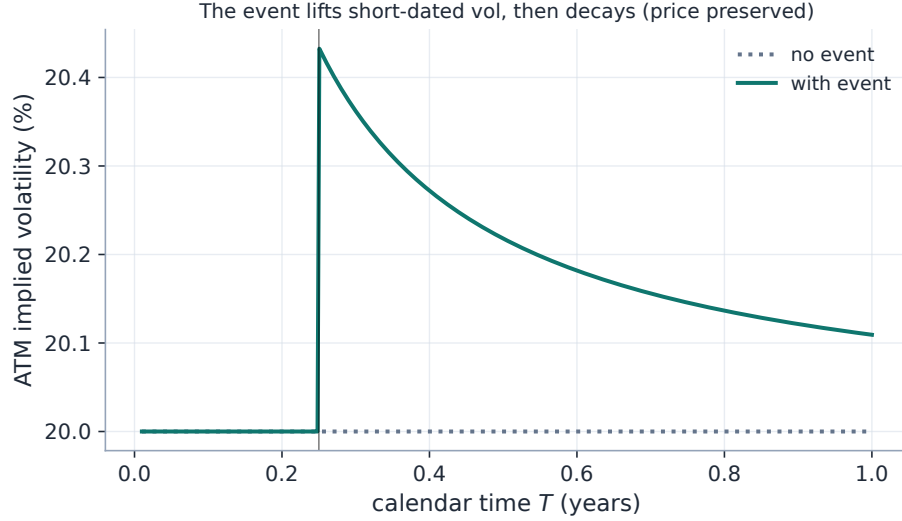


Figure 2: ATM implied-vol term structure with and without the event. The event lifts short-dated vol and decays away; the option *prices* are preserved, only the implied-vol reading changes.

2.3 Normalization

By default the cumulative weight exceeds the calendar-day count ($\tau > t$). With normalization on (an Options toggle), all days — event days included — are rescaled by a single factor so the one-year weight budget stays 365:

$$\tau_{\text{days}}(t) \mapsto \tau_{\text{days}}(t) \cdot \frac{365}{365 + \sum_{t_e \leq 1} N_e}. \quad (3)$$

Then the one-year variance time — and the one-year implied vol — are identical to a no-event year: events only *redistribute* variance within the year, they do not add it. Which convention to use is a desk choice: un-normalized treats the events as genuinely extra variance, normalized treats them as a known reallocation of a fixed annual budget.

3 How it threads the pipeline

The dual clock is prepared once per node by the clock of listing 1: τ is stored as `prepared.tau` beside the calendar `prepared.t`, and every fit (parametric and local-vol), the local-vol grid, the term structure and the option table consume τ in place of t .

Remark 1 (One production clock, one legacy clock, one unit trap). Two dilation clocks exist in the codebase and must not be conflated. The production clock is `calib/weighted_time.py` — day-weighted, exactly (1), the one every fit consumes. The older `calib/event_time.py` (`EventClock`) survives only for term-structure interpolation. And the unit trap: `weight` is *extra equivalent days*, never years — confirmed both by the call site (the pairs feed (1) directly) and by the auto-calibrator, which constructs `EventSpec(weight=days)`. The `EventSpec` schema docstring said *years* until 2026-07-09 (now fixed to the day semantics documented here); the legacy clock’s comments remain the stale party.

Total variance is interpolated *linearly in τ* between quoted nodes — flat forward variance per weighted-time unit — so the dense vol curve $\sqrt{w/\tau}$ stays bounded near zero and sensible past the

last expiry. The term-structure view uses the shared calendar (not the per-node events) so the displayed curve is coherent across expiries. Changing the event calendar bumps the per-ticker events version, forcing a refit of only that name; with an empty calendar the whole pipeline is byte-identical to calendar time.

Nor must the calendar be hand-entered. The Term workspace’s *auto-calibrate* action infers it from the ATM term structure: placing one candidate event before each expiry up to a chosen horizon (`maxExpiry`), it solves for the non-negative extra days N_e that make the *event-time* forward variance between consecutive expiries, $(w_i - w_{i-1})/(\tau_i - \tau_{i-1})$, as flat and as monotone-increasing as possible while keeping the events small and sparse (a bounded L-BFGS-B solve; events under half a day are thresholded away, so the result is a short list). The calendar-time forward variance $(w_i - w_{i-1})/(t_i - t_{i-1})$ is *event-invariant* — prices fix w — so the objective works on the dilated clock, the one events actually move: since an event can only *lengthen* an interval’s weighted time, it pulls *down* the forward-variance spike an earnings-packed interval produces, toward its neighbours. The calendar is thus read off exactly the kinks the clock was built to explain, installed as the shared per-ticker calendar, and edited like any hand-entered one (`calib/event_autocalib.py`, `api/event_autocalib.py`).

4 What is original here

Event time changes are not new in spirit (Bergomi and others dilate time around events), but the implementation here has two clean properties worth singling out. First, *price preservation by construction*: the event clock never touches $w(k)$, only the volatility read-off (2), so adding an event can never change an option price — it cannot introduce arbitrage. Second, the *day-weighted* parametrization (1) makes the event size a single intuitive number (extra equivalent days) that a desk can quote and reason about, with an optional budget normalization (3) that cleanly separates “extra variance” from “reallocated variance”. And the byte-identical fallback when the calendar is empty means the feature is strictly additive. The payoff is comparability: pre- and post-earnings surfaces line up across names and dates because event-time vol is the signal — calendar-time vol is the artifact.

5 Traceability

Table 1: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
No events: identity clock, byte-identical pipeline	invariant 2	<code>calib/weighted_time.py</code>	<code>test_weighted_time.py::test_no_events_is_identity, ::test_no_event_byte_identical</code>
Event adds day weights before its date only	(1)	<code>calib/weighted_time.py</code>	<code>test_weighted_time.py::test_event_before_adds_day_weights</code>

Claim	Object	Code anchor	Test anchor
Price preservation: IV drops by $\sqrt{t/\tau}$ (2)		api/service.py, api/ displayed.py	test_weighted_time.py:: test_event_lowers_iv_ by_sqrt_t_over_tau, ::test_localvol_iv_drops_ with_event
Normalization pins the one-year bud- get (3)		calib/weighted_time. py	test_weighted_time.py:: test_normalization_ pins_one_year, ::test_ normalization_keeps_one_ year_vol
Surface mesh shares the smile’s clock section 3		api/affine_views.py	test_surface_clock.py:: test_surface_mesh_uses_ variance_clock_like_the_ smile
Legacy clock confined to term inter- polation remark		calib/event_time.py	test_event_time.py:: test_interpolation_lumps_ event_variance
Auto-calibrate flattens event-time for- ward variance; flat input stays event- less section 3		calib/event_ autocalib.py, api/ event_autocalib.py	test_event_autocalib.py:: test_spike_is_flattened, ::test_flat_input_stays_ eventless

A Hyperparameter atlas

Table 2: Event-clock hyperparameters.

Knob	Default	Role
<i>Surfaced (OptionsSettings)</i>		
events	empty	List of EventSpec (t_e, N_e, label); N_e = extra equivalent days.
normalizeEvents	off	Rescale all days to a fixed 365 one-year budget (3).
maxExpiry	—	Auto-calibrate horizon: one candidate event per expiry at or before it (section 3).
<i>Hidden</i>		
DAYS_PER_YEAR	365	Calendar-to-variance day conversion.
mono_weight	1.0	Auto-calibrate: penalty on decreasing event-time forward variance.
sparse_weight	/ 10^{-3} / 10^{-4}	Auto-calibrate: keep solved events small and sparse.
ridge_weight		
min_event_days	0.5	Auto-calibrate sparsity threshold: smaller solved events are dropped.

Performance. The clock is a scalar sum per node, negligible. With no events $\tau = t$ and the entire

downstream pipeline is byte-identical, so the feature adds zero cost when unused; changing the event calendar refits only the affected ticker (per-ticker events version).

References

- [1] L. Bergomi. *Stochastic Volatility Modeling*. CRC Press, 2016.
- [2] J. Gatheral. *The Volatility Surface*. Wiley, 2006.